

D25010001_DIT1083_DIT Mohamed Aiman

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DIPLOMA IN INFORMATION TECHNOLOGY

FINAL ASSESSMENT (CASE STUDY)

ANSWER BOOKLET

COURSE	:	DATA COMMUNICATION & NETWORKING
COURSE CODE	:	DIT1083
DATE	:	13 th AUGUST 2026
TIME	:	3.00 PM – 9.00 PM
DURATION	:	6 HOURS

INSTRUCTIONS TO CANDIDATES:

1. This case study consists of **TWO Sections**. Answer all sections.
2. All answers are to be written in MS word in the answer script booklet given and convert the file into PDF format. Submit the link provided on the BLMS.
3. Remember to save your document. Use your ID number and Course Code as title for the folder.
Example: DBA2410006_DBA1044
4. You have **SIX (6)** hours to submit your answers.
5. Any submission with a Turnitin similarity index exceeding the prescribed threshold (30% for undergraduate-level work and 20% for postgraduate-level work), as stipulated in the Guidelines for the Responsible and Ethical Use of Artificial Intelligence (AI) Technologies (MUST/QA/ETHICAL-AI-USE/02), is not acceptable.

STUDENT'S ID NO. : Mohamed Aiman D25010001

LECTURER : NURULAINI BT SHARIFFUDIN

COURSE LEARNING OUTCOMES

CLO 1	Explain of basics computer networking.(C2, PLO1)	PLO1
CLO 2	Differentiate topologies of network (C4, PLO5)	PLO5
CLO 3	Discover the emerging networking trends (C3,PLO9)	PLO9

2 STATE THE CLO/PLO		ATTAINMENT OF AoL
Section 1-2	2 CLO 3: PLO 9	Cognitive Skills

* Competency Skill in Study: Knowledge and Understanding (KU), Cognitive Skills (CS), Interpersonal Skills (IS), Practical Skills (PS), Ethic and Professionalism (EP), Numeracy Skills (NS), Digital Skills (DS)

**This page is for moderator's and examiner's reference only.

Submission Requirements:

- 5 • Use font size 12 and single line spacing.
- All submissions must include proper referencing using either APA referencing style.

QUESTION 1**CASE STUDY A****Introduction**

Sunrise College is outgrown a network that seems to have been designed for a smaller campus. The symptoms are not individual minor hassles. Slow access, unreliable file sharing, dropped internet lessons, or low traffic separation, and possibly weak internet edge, are signs of congestion, low wireless capacity, or capacity, lack of switching, poor wireless coverage, and/or undersized internet edge. A coordinated network of the campus, then, is the proper answer, one that isn't just a faster router.

a)

Computer networking is the process of linking together computer systems using physical connections (wired or wireless) and a defined communication protocol to share resources and transmit data between systems. These devices range from computers, servers, printers, phones, access points, switches to routers. A network is, in essence, a way to make separate devices an information system that can be coordinated. Computer networking is also defined by AWS (n.d.) as a system of interconnected devices, resource sharing, links, nodes and communication rules.

Sunrise College's networking is the virtual glue that holds the existing campus, new Academic Block, digital library and internet-based services together. A good LAN setup might enable some 700 people to get access to the LMS, electronic journals, shared drives, printers, and administrative databases, without being required to carry files on removable media. According to Cisco (n.d. -a), a LAN can be as small as a home network, and as large as an enterprise or a school network that has thousands of users, but is defined by the fact that it's limited physically.

It is also of operational importance. Sensitivity to latency, jitter and packet loss are important for online classes. Short disconnections are also problematic. A high throughput and reliable permissions are required for file sharing. Student records must be kept confidential and guests should be given access to separate internet systems to the finance and academic systems. These various needs can be met through the use of a modern network, which includes switched Ethernet, managed Wi-Fi, virtual LANs, quality of service, authentication, monitoring, and resilient links. The real benefit of VLANs is that they can partition a physical switched infrastructure into logical groups, minimize unnecessary broadcast traffic, and isolate users with high security needs, such as Cisco (Cisco, 2026).

(10 marks)

b)

Resource sharing and reduced duplication. Users can share Internet access, network printers, licensed applications, digital library databases and central storage. According to Cisco (n.d.-a), key LAN benefits include shared internet, files and printers. For the college, central resources mean repeated purchases are avoided, enabling the IT staff to maintain one controlled service instead of multiple copies.

Pronges faster collaboration and communication. Shared documents, group project folders, messaging, email, voice and video meetings are all on a single network. The same project resources can be accessed by students from the classroom and in the library, and the lecturer can share materials immediately. Version control for cloud or server folders also eliminates the confusion of multiple copies of files.

Centralized management, security and backup. Consistent user management, permissions, software updates, anti-malware protection, and back-ups are possible. Students, staff, administration, server, guest, voice and building devices are separated onto different VLANs, which eliminates unnecessary exposure. This benefits least-privilege access, which is consistent with the concept of zero-trust that access should only be granted when a device is on the internal network (Rose et al., 2020).

Mobility and dependable access to learning. Students and staff can walk from the academic block to the digital library without losing their wireless connection. A WLAN can be used to connect different devices and can be managed from a single software interface, making it easier to grow and monitor (Cisco, n.d.-e). Access points also help to minimize some dead zones and overloaded "hotspots" that interrupt lessons when used properly.

(10 marks)

c)

A LAN and a WAN both move data, but they solve different geographical and organisational problems. Sunrise College needs a high-speed campus LAN internally and a WAN or internet service at the edge to reach cloud platforms, external websites, remote campuses, or off-site users.

Table 1. LAN and WAN differences applied to Sunrise College

Criterion	LAN	WAN	Sunrise College application
Coverage area	A room, building, or limited campus area.	A city, country, region, or global area. It may connect several LANs.	The academic block, library, offices, laboratories, and campus Wi-Fi form the LAN. Internet and any remote-site links form the WAN.
Ownership and control	Usually owned or directly controlled by one organisation, including its switches, cabling, access points, addressing, and policies.	Usually depends partly on telecommunications carriers or internet service providers. The customer controls its routers and service contract, but not the entire carrier network.	College IT controls VLANs, Wi-Fi, switches, and servers. An ISP controls the external transmission service.
Typical application	Fast access to nearby servers, printers, storage,	Cloud access, inter-campus communication, remote access, email delivery, and	Local files should remain fast even if the internet is busy, while the WAN carries

	laboratories, and internal systems.	internet connectivity over distance.	LMS cloud traffic, video calls, and external research access.
Performance and cost pattern	Typically high throughput and low delay per bit because distances are short and infrastructure is locally managed.	Performance depends on contracted bandwidth, distance, congestion, provider design, and service level. Recurring cost is normally higher.	Use multi-gigabit access where justified, 10 Gb/s fibre uplinks between blocks, and a business-grade internet service sized from measured peak demand.

Cisco (n.d.-a) describes a LAN as devices connected in one limited physical location and contrasts it with WANs that cover larger geographic areas and connect multiple LANs. Therefore, the terms should not be treated as competing choices. They are complementary layers of the college's end-to-end service.

(10 marks)

d)

Table 2. Device recommendations and case-specific functions

Device	Main function	Recommended use at Sunrise College
Managed Layer 3 PoE switches	Forward Ethernet frames inside the LAN, create VLANs, prioritise traffic, aggregate links, and route between approved VLANs. Power over Ethernet can supply access points and IP phones.	Use access switches on each floor and a redundant or stacked core. Provide fibre uplinks between buildings, sufficient port capacity, link aggregation, monitoring, and spare ports for expansion.
Business edge router with next-generation firewall	Connect the campus LAN to the WAN and select routes. The security gateway should also enforce firewall rules, network address translation, VPN access, content controls, intrusion prevention, and application-aware policies.	Deploy at the internet edge with throughput that remains adequate when security inspection is enabled. Prefer dual power and, if budget allows, a second ISP or cellular failover for critical online learning.
Controller-managed Wi-Fi 6 access points	Bridge wireless devices into the wired LAN, provide radio coverage, support secure authentication, and share available airtime among many clients.	Conduct a wireless site survey, then place APs according to capacity and signal conditions in lecture halls, library spaces, offices, and corridors. Use separate secure staff, student, and guest SSIDs mapped to VLANs.

Switches, routers, and wireless access points are distinct but complementary foundations of a business network (Cisco, n.d.-b). Wi-Fi 6 is appropriate in dense learning areas because OFDMA divides channels into subcarriers and can improve throughput, latency, and service to many users (Cisco, n.d.-c). The APs still need a strong wired switching and fibre backbone, since wireless traffic ultimately enters the wired network.

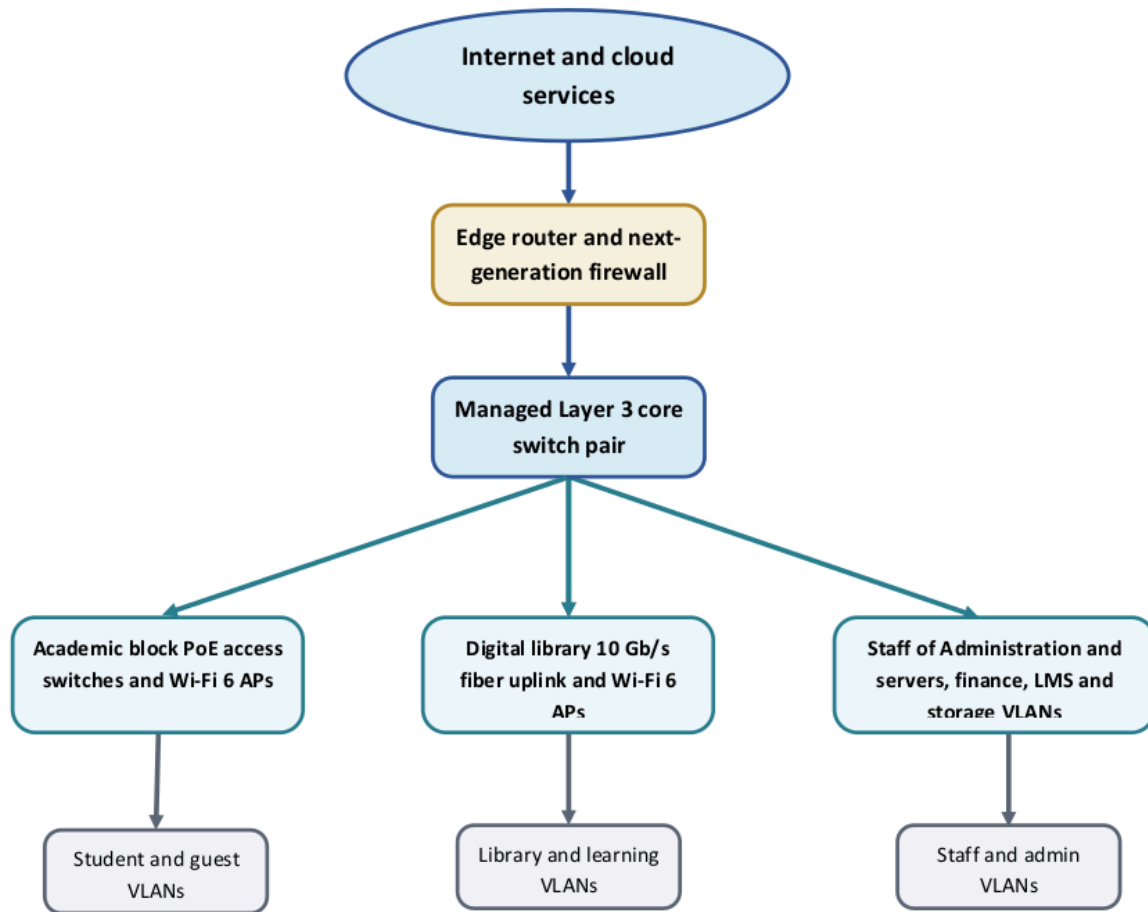


Figure 1. Proposed Sunrise College network architecture

Figure 1 uses a hierarchical star arrangement. The design separates users logically while retaining controlled routing through the core. The final number and placement of access points should be determined by a predictive design followed by an on-site validation survey, not by floor area alone.

(10 marks)

e)

Lecture capture, demonstrations, video conferencing and cloud learning platforms are essential to enhancing teaching and learning, which is improved by predictable service. Quality of Service may favour voice and video over large downloads as background traffic, and capacity-based Wi-Fi design minimises the competition in lecture hall spaces. Lecturers can easily go from room to room, access teaching materials in a timely fashion, and conduct blended classes with less interruptions.

Learning is more inclusive and collaborative. Students can access the e-books, journals, virtual laboratories, recorded classes, and collaborative project areas in the academic block or library. Reliable access also means reduced downtime for reconnections, waiting for downloads or manually transferring files. Therefore, the network is not only fast, but it is also continuous, a feature which is necessary for real-time lessons and timed assessments.

Central records, secure workflows of files, networked printing, backup, and uniform access control are all advantages for administrative work. The admissions, finance, human resources and examination team can share authorised information without having sensitive systems on the same open segment as guests. Monitoring data can alert to congestion, dead links, rogue devices and unusual traffic patterns early. The stages of a staged implementation should consist of baseline measurements, VLAN and addressing design, cabling and wireless surveys, pilot testing, user communication and post-installation performance checks.

Governance of technology is thus the prime improvement. The college should establish goals for service in terms of availability, wireless coverage, peak utilisation, video quality, backup recovery and incident response. This turns the upgrade into an actual learning service that can be measured.

(10 marks)

[50 marks]

QUESTION 2

CASE STUDY B

Introduction

A new office is needed for techVision that will house 80 employees, and grow to accommodate 150 employees in three years. It is the most demanding services, cloud applications, shared files, video conferences, which demand stable access, quick fault isolation and easy expandability. Physical topology should attempt, therefore, to reduce shared failure points in an economical manner to cable and manage.

a)

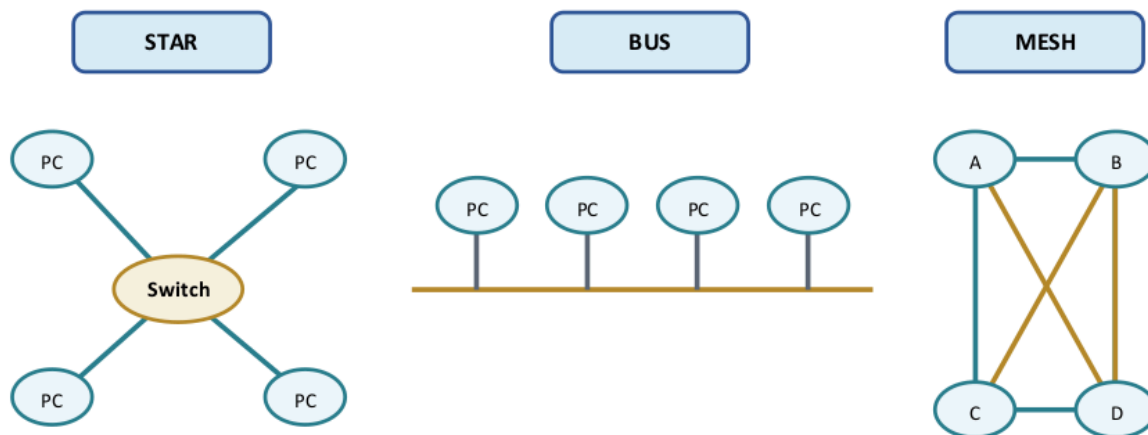


Figure 2. Star, bus, and full-mesh topologies

A star topology has a single switch plugged into each end-point. Traffic between endpoints is forwarded through that switch. When an endpoint cable fails, it is likely to impact only one user and new endpoint devices can be added to available switch ports. Central switch is significant, though, as it is the switch that failure of that switch can disrupt all of the users attached to it, unless the design includes stacking, redundant core switches, spare equipment, and diverse uplinks.

A bus topology is a type of topology in which devices are connected to a single backbone medium. The frames run on the common cable and used to have to be terminated at both ends. Requires limited amount of cable and no central switch, it does not implement collision detection as older Ethernet standards do, is hard to locate a fault, and a fault in the backbone affects all other segments of the network; not suitable for a modern 80-person office. According to AWS (n.d.) topology refers to the configuration of nodes and links, with the three types of topologies determined by the way the links are organised, namely as a bus, star, or mesh.

Mesh topology: Nodes are connected to each other in many ways. In a full mesh, all nodes are connected directly to each other; in a partial mesh just selected critical nodes have multiple connections. The full mesh has good path redundancy and will grow rapidly: There are $n(n - 1)/2$ links in the mesh, but they are undirected. The number of endpoints would be 80, so the number of links needed would be 3,160, which is economically and operationally impractical. Partial mesh is useful when a few key switches,

gateways or locations are critical and redundancy is required with the additional links.

(10 marks)

b)

Table 3. Cost, reliability, and maintenance comparison

Topology	Cost	Reliability	Maintenance	Overall judgement for TechVision
Star	Moderate. More cable and a switch are required, but structured cabling and commodity Ethernet keep costs predictable.	Good. One access link failure affects one user. The central device remains a risk unless made redundant.	Easy. Ports can be monitored, faults isolated, and devices added without disturbing the whole network.	Best balance for office access.
Bus	Low initial hardware and cable cost, but legacy components and downtime raise lifecycle cost.	Low. A backbone or termination fault can affect many users, and all devices share the medium.	Difficult. Faults and performance problems are harder to isolate; expansion can disrupt service.	Not suitable for modern office Ethernet.
Full mesh	Very high. Links, interfaces, installation, and configuration rise rapidly with each node.	Very high path redundancy when routing can use alternate links.	Complex. Many links, routes, and failure states must be monitored and documented.	Use only as a limited partial mesh in the core or WAN, not between employee devices.

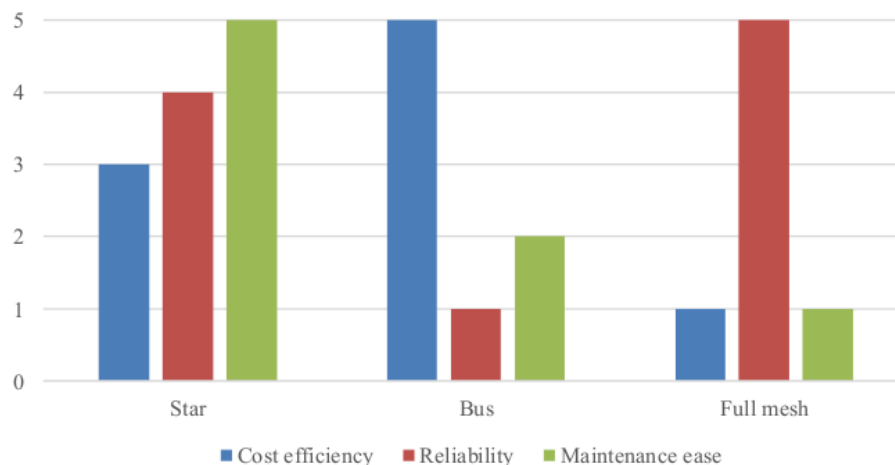


Figure 3. Topology suitability factors, where 5 is most favourable

The chart is not actual experimental data, but a clear case analysis score. The higher the cost score, the more efficient the cost. The image demonstrates why star, the best compromise. Bus is inexpensive, but with poor reliability, and full mesh is reliable but too expensive and complex. AWS also advises that a many-to-many mesh architecture will have higher infrastructure and management costs, and the hub-and-spoke model will be easier to manage and operate (Amazon Web Services, 2025).

(10 marks)

c)

Redundancy is needed at the center of the network, and this should be done using a switched extended-star topology. All employees' devices, access points, printers, and IP phones are plugged into access switches. The access switches connect via high-speed uplinks to a stacked or paired core. If cloud and conferencing availability are business critical, the two security gateways and internet links should be redundant, too. This design retains clarity of a star design but only a partial mesh is used where one failure would impact the company.

- Performance: Dedicated switched links avoid the shared-medium limitations of a bus. Multi-gigabit uplinks and quality of service can prioritise conferencing, voice, and interactive cloud sessions.
- Reliability: A failed endpoint cable affects one employee. Redundant core switches, gateway failover, link aggregation, uninterruptible power, and diverse internet services reduce central failure risk.
- Maintenance: The IT team can identify errors by switch port, monitor utilisation, replace one access switch, and expand one floor without recabling the whole office.
- Security: VLANs can separate corporate users, servers, voice, guests, building systems, and management traffic. Access policies then control permitted paths. This is more defensible than trusting every device merely because it is inside the office, consistent with zero-trust architecture (Rose et al., 2020).
- Growth: Standard 48-port PoE access switches, spare fibre pairs, rack space, power, IP address capacity, and controller licences can be added in repeatable blocks.

(10 marks)

d)

Trend 1: Software-defined networking and intent-based automation

Software-defined networking (SDN) decouples or abstracts the network control from the packet-forwarding functions, and concentrates network policies into a controller. Cisco (n.d.-d) says that SDN offers more flexibility and ease of network management, and is implemented via centralised control, automation, and programmable interfaces. This can be deployed via cloud-managed or on-premises controllers on switches, access points and gateways.

Consistency is the key to the practical value. A new employee or department can be assigned to a VLAN, access policy and quality of service profile using a new employee or department template. The controller is able to gather telemetry, report on client experience, identify abnormal traffic, back up client configuration, and report on application using the bandwidth. With a video problem, the IT staff are able to spot internet congestion, Wi-Fi interference, packet loss or an overloaded uplink faster. Nonetheless, change approval, role-based administration, secure APIs and configuration backups need to be in place to ensure that automation is governed.

Trend 2: Wi-Fi 6 high-density wireless networking

The Wi-Fi 6 standard is for areas with lots of devices operating concurrently. OFDMA can partition a channel into smaller resource units (RU); a way to improve the ability of an access point to provide service to multiple clients. According to Cisco (n.d.-c), OFDMA is correlated with increased throughput and reduced latency and enhanced performance in dense areas. Complimentary to this is the MU-MIMO feature which enables the suitable access points to send and receive data to multiple devices through multiple spatial streams (Cisco, n.d.-f). Retaining wired Ethernet for fixed workstations and bandwidth-heavy devices, while using controller-managed Wi-Fi 6 APs for laptops, phones, collaboration spaces and guest access offers the perfect balance between flexibility and performance.

By combining controller-managed Wi-Fi 6 APs with laptops, phones, collaboration spaces, and guest access with wired Ethernet for fixed workstations and bandwidth-heavy devices, TechVision provides the best of both worlds. The placement of APs, channel widths, power levels, and capacity should be based on a site survey. Use enterprise authentication for corporate access, isolate guest access, and remove old encryption. It's not just about the advertised speed. It is more efficient, better behaviour under load and easier scaling with increasing number of devices and employees.

(10 marks)

e)

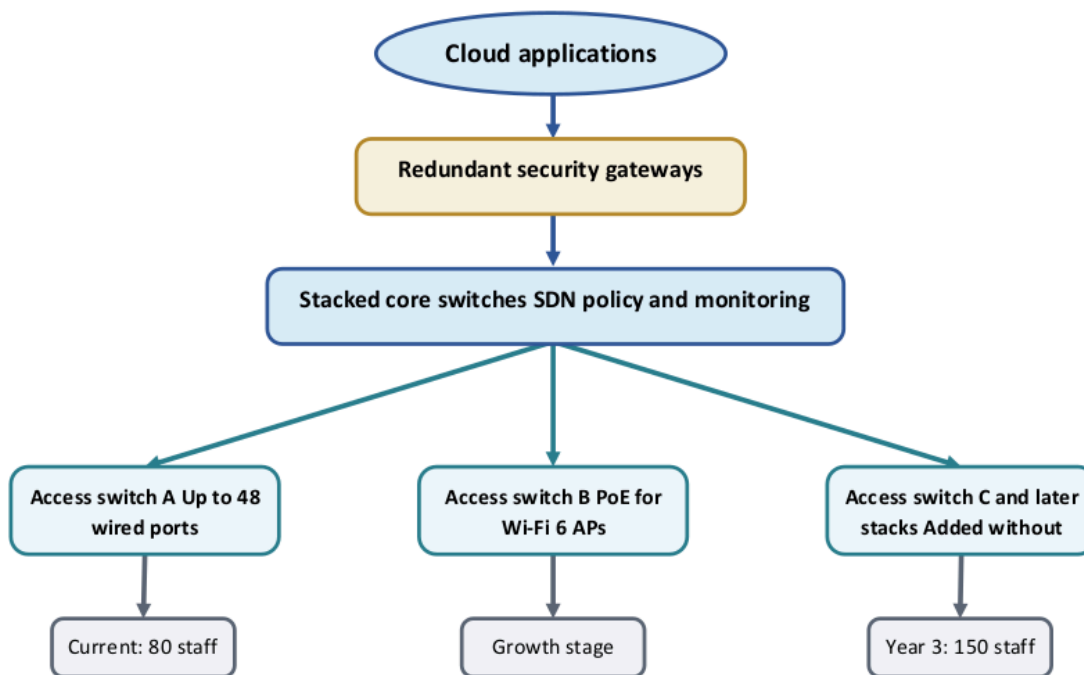


Figure 4. Scalable extended-star design for TechVision

The suggested extended star scales are those in Modules. TechVision can deploy enough 48-port access switches for existing wired end points, APs, phones, printers, meeting-room systems, and provide a minimum of 25 percent margin. As the recruitment continues, additional APs and access switches can be incorporated into the same structured cabling, core, monitoring and policy. The design should provide rack units, power, cooling, fibre strands, switch stack capacity, IP subnet, DHCP scope, software licences, and 150 people as the design target.

The core should be chosen for the future not the first day. The redundant high speed uplinks ensure that one access switch is not rendered an island, and link aggregation ensures a high capacity and resilience. Internet sizing should take into account real 95th-percentile usage, simultaneous video sessions, cloud application consumption, backup periods, and service-level goals. If the loss of the internet connection would prevent the majority of employees from being able to perform their duties, then a secondary internet path is warranted.

SDN enables growth through default policies and configurations per new switch, AP, user group, branch. Central telemetry allows users to see capacity trends before users experience congestion. While Wi-Fi 6 enables a growing number of mobile devices to connect via more efficient channel utilisation, TechVision must also include AP capacity, not just through increasing transmit power, but by basing it on actual clients per radio and airtime utilisation.

Security should grow in proportion to the network. Expansion should be accompanied by identity-based access, multifactor authentication for sensitive services, endpoint health checks, segmented guest access, logging and regular policy reviews. In contrast, NIST says that zero trust is focused on authenticating and authorising explicitly, not implicitly, based on network location (Rose et al., 2020). It is particularly relevant given the advent of cloud services, remote working, contractors and increased devices at TechVision.

Last but not least, relying on clouds should be recognised as a capacity and continuity issue. Cloud computing is built on the principles of broad network access and fast elasticity (Mell & Grance, 2011), which rely on a good office network and Internet connection. The star topology, SDN automation and Wi-Fi 6 combined provide TechVision with a controlled expansion without the network redesign through every recruitment phase.

(10 marks)

Conclusion

In both examples, network quality is the result of the combination of architecture, capacity planning, security, and operations. Sunrise College requires a segmented campus LAN with managed switching, secure WAN edge, Wi-Fi 6 connectivity and fibre connectivity between buildings. For TechVision, they require a long, redundant, SDM, high-density wireless access star. For both organisations, success can be measured by user experience, availability, latency, loss, usage, incident recovery and how easily authorised users can access the resources that are important to them.

[50 marks]

[Total 100 marks]

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